Toward a Next-Generation First-Principles Theory of Electron–Phonon Coupling

Why DFPT Works, Where It Fails, and a Channel-Resolved Correction Strategy

AITP Research Notes

30 July 2026

Abstract

Density-functional perturbation theory (DFPT) is the standard first-principles method for lattice dynamics and electron–phonon coupling. Its practical success extends from simple metals and semiconductors to several materials with appreciable electronic correlations, even though the physical scattering amplitude is a many-body vertex and DFPT uses a static Kohn–Sham response. The central question of this proposal is therefore not whether corrections to DFPT exist, but why they are often small for some phonons and large for others.

We formulate a testable working hypothesis: the DFPT error is controlled primarily by the operator channel in which a phonon perturbs the low-energy electronic Hamiltonian, together with the channel-resolved many-body susceptibility and its frequency dependence. A common charge-like perturbation is constrained in the strict uniform limit and has a particularly simple realization in the uniform electron gas. Orbital splitting, hybridization, hopping, spin, and interaction modulation are not protected by the same constraint. This distinction explains why strong correlation alone is not a sufficient failure criterion and motivates a next-generation method: preserve DFPT as a low-cost baseline, project each phonon perturbation into physically defined channels, and add only the static or dynamic many-body corrections required by those channels. The proposal specifies the equations, limits, benchmark materials, falsification criteria, a two-day literature sprint, and an AI-agent workflow for turning the hypothesis into a transparent prediction framework.

Proposed deliverable. The intended result is not a universal scalar multiplier for every electron–phonon matrix element. It is a channel-resolved, uncertainty-aware correction layer on top of DFPT that can also decide when uncorrected DFPT is already adequate.

Contents

List of Figures

List of Tables

1 Problem statement and scope

In crystalline solids, the interaction between electrons and lattice vibrations controls electrical resistance, conventional superconductivity, thermoelectric transport, carrier relaxation, and device heating. DFPT is the most widely used first-principles tool for calculating phonons and electron–phonon matrix elements because it avoids a separate large-supercell calculation for every displacement and treats the self-consistent electronic response directly [Baroni et al., 2001, Gonze, 1995, Giustino, 2017].

The motivating observation of Quantum Harness Issue 35 is that DFPT is often quantitatively useful in a surprisingly broad range of materials [QuantumBFS, 2026]. This statement needs careful wording. It does *not* mean that ordinary semilocal DFPT is uniformly accurate for every correlated material, every phonon, and every observable. Standard DFPT can fail when the reference electronic structure is qualitatively wrong, when a phonon couples to an unprotected internal degree of freedom, or when the electronic response is strongly frequency dependent. The real puzzle is narrower and more useful:

Core scientific question. Why are corrections beyond static Kohn–Sham linear response small for some electron–phonon perturbations, including some perturbations in correlated materials, while they are large for other modes in the same material?

This formulation rules out two incomplete answers.

1. **"The quasiparticle residue cancels the vertex."** A numerical product close to one is evidence, not a physical explanation. The Ward identity produces such a cancellation only in a specific conserved-charge limit, whose order of limits is not the generic adiabatic phonon limit.
2. **"DFPT works because the phonon shifts every electronic state by the same amount."** Real perturbations do not act equally on all Bloch states. Even a local operator proportional to the identity inside a correlated orbital subspace produces band- and momentum-dependent matrix elements because Bloch states have different orbital weights and phases.

The proposal instead treats the one-body derivative of the electronic Hamiltonian as an operator. It separates the single global identity `global_charge`, relative block-identity shifts `site_charge`, block-internal traceless structure `internal`, and inter-block matrix elements `nonlocal`. Derivatives of interaction parameters are two-body vertices in a different operator space. Conservation laws can constrain only the first one-body direction, and only when the original unprojected perturbation is independently verified as a strict uniform $\boldsymbol{q} = 0$ full-space common shift.

1.1 Target deliverable

The target is a computational framework that uses DFPT as a baseline and adds a small number of physically interpretable, channel-resolved many-body corrections. It should be:

- **simpler** than a full many-body calculation of every $g_{mn\nu}(\boldsymbol{k}, \boldsymbol{q})$;
- **potentially cheaper than a dense beyond-DFPT vertex grid** by focusing expensive calculations on risky modes and low-dimensional operator channels, without claiming a measured speedup over DFPT;

- **transparent** because every correction is attached to a named physical operator and a documented validity test;
- **falsifiable** on materials for which DFPT succeeds for one mode but fails for another; and
- **uncertainty-aware**, including an abstention path when a mode lies outside the validated regime.

The first validation object may be the uniform electron gas (UEG), but it is not the final theory. The UEG contains only a scalar density channel and therefore provides a clean baseline for understanding what is special about that channel. The proposed method must then survive multi-orbital and correlated-material tests.

1.2 What is and is not being corrected

DFPT computes several related quantities: force constants, phonon eigenvectors, Born effective charges, dielectric response, and electron–phonon matrix elements. The main target here is the last quantity: the physical quasiparticle scattering vertex associated with a phonon. Corrections to equilibrium structure, anharmonicity, or the phonon spectrum may also matter, but they are separate axes of error. The proposed benchmark will keep the following gates distinct:

1. the Kohn–Sham ground-state and phonon baseline;
2. the electron–phonon matrix element g;
3. the observable assembled from g, such as a linewidth, mobility, or pairing kernel.

Passing one gate does not establish the next.

2 What standard DFPT actually assumes

2.1 The perturbation and the matrix element

Let $u_{\nu\boldsymbol{q}}$ be the normal coordinate of phonon branch ν and wavevector $\boldsymbol{q}$. After including the conventional phonon normalization factor, define the self-consistent Kohn–Sham perturbation operator

$$D^{\mathrm{KS}}_{\nu\boldsymbol{q}} \equiv \partial_{\nu\boldsymbol{q}} V_{\mathrm{KS}} = \partial_{\nu\boldsymbol{q}} V_{\mathrm{ion}} + \partial_{\nu\boldsymbol{q}} V_{\mathrm{Hxc}}. \tag{1}$$

The DFPT electron–phonon matrix element is

$$g^{\mathrm{DFPT}}_{mn\nu}(\boldsymbol{k},\boldsymbol{q}) = \langle \psi_{m,\boldsymbol{k}+\boldsymbol{q}} | D^{\mathrm{KS}}_{\nu\boldsymbol{q}} | \psi_{n,\boldsymbol{k}} \rangle. \tag{2}$$

The perturbation is a matrix in band, orbital, spin, and reciprocal-lattice indices. Therefore standard DFPT does not assume that every electron sees the same scalar shift. For a general Fourier component the operator that transfers $\boldsymbol{k}$ to $\boldsymbol{k}+\boldsymbol{q}$ is not Hermitian by itself; a real lattice displacement gives

$$\left(D^{\mathrm{KS}}_{\nu\boldsymbol{q}}\right)^{\dagger} = D^{\mathrm{KS}}_{\nu,-\boldsymbol{q}}. \tag{3}$$

The executable Hermitian prototype below is therefore restricted to a real-space displacement derivative, a Gamma-point perturbation, or a real standing-wave combination of $\boldsymbol{q}$ and $-\boldsymbol{q}$. It does not expose a general continuous fixed-$\boldsymbol{q}$ matrix interface.

The first-order orbital response is obtained from a Sternheimer equation of the schematic form

$$\big(\hat{H}_{\mathrm{KS}} - \varepsilon_{n\boldsymbol{k}}\big)|\partial_{\nu\boldsymbol{q}}\psi_{n\boldsymbol{k}}\rangle = -\hat{Q}_{\boldsymbol{k}+\boldsymbol{q}}\big(\partial_{\nu\boldsymbol{q}} V_{\mathrm{KS}} - \partial_{\nu\boldsymbol{q}}\varepsilon_{n\boldsymbol{k}}\big)|\psi_{n\boldsymbol{k}}\rangle, \tag{4}$$

where $\hat{Q}$ projects outside the occupied state or selected variational subspace. The induced density updates V_{Hxc} until self-consistency.

2.2 Static electronic equilibrium

In ordinary adiabatic DFPT, the electrons are assumed to remain in the instantaneous ground state associated with a slowly displaced lattice [Gonze, 1995, Baroni et al., 2001]. Equivalently, the electronic response entering equation (4) is evaluated at zero external frequency. The electronic denominator contains Kohn–Sham energy differences, but not a phonon-frequency shift such as $\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{n\boldsymbol{k}} \pm \omega_{\nu\boldsymbol{q}}$.

This assumption is usually sensible when electronic relaxation and interband energy scales are fast compared with the phonon period. It can fail for low carrier densities, narrow bands, small Fermi energies, polar optical modes, collective modes, or any situation in which $\omega_{\nu\boldsymbol{q}}$ is not negligible on the scale governing the electronic response.

2.3 The response equation behind DFPT screening

For a scalar density perturbation, suppressing matrix indices for clarity,

$$\delta V_{\mathrm{KS}} = \delta V_{\mathrm{ion}} + (v + f_{\mathrm{xc}})\delta n, \tag{5}$$

$$\delta n = \chi_s \, \delta V_{\mathrm{KS}}, \tag{6}$$

where v is the Coulomb interaction, f_{xc} is the static exchange–correlation kernel, and χ_s is the independent Kohn–Sham density response. Solving the two equations gives

$$\delta V_{\mathrm{KS}} = \left[1 - (v + f_{\mathrm{xc}})\chi_s\right]^{-1} \delta V_{\mathrm{ion}}, \qquad g^{\mathrm{DFPT}} = \frac{g^0}{1 - (v + f_{\mathrm{xc}})\chi_s} \tag{7}$$

in a translationally invariant scalar problem. In a crystal these quantities are matrices in reciprocal lattice vectors and may also carry spin or sublattice indices.

2.4 The precise missing term

Choose the convention

$$G^{-1}(\omega) = \omega + \mu - H_{\mathrm{KS}} - \Sigma^{\mathrm{corr}}(\omega), \tag{8}$$

where Σ^{corr} is the self-energy correction beyond the chosen Kohn–Sham reference, with the corresponding double counting removed. The fixed-basis one-body inverse-Green-function vertex is

$$\Lambda^{(1)}_{\nu\boldsymbol{q}}(\omega) \equiv -\partial_{\nu\boldsymbol{q}} G^{-1}(\omega) = \partial_{\nu\boldsymbol{q}}\left[V_{\mathrm{KS}} + \Sigma^{\mathrm{corr}}(\omega)\right], \tag{9}$$

Thus the central vertex term absent from standard DFPT is $\partial_{\nu\boldsymbol{q}}\Sigma^{\mathrm{corr}}$, together with its frequency dependence. Quasiparticle external-leg factors and state rotations are separate from the fixed-basis vertex in equation (9) and must be applied consistently when constructing an observable. A large equilibrium self-energy does not logically imply a large derivative with respect to a particular phonon. This distinction is the first reason why the label “strongly correlated” alone cannot predict the DFPT error. A consistent separation between the DFT starting point and many-body electron–phonon corrections is discussed in detail by Marini et al. [2015].

Operational limitation of DFPT. Standard DFPT is a static Kohn–Sham response. Its reliability for a physical quasiparticle vertex depends on whether the phonon derivative of the beyond-DFT self-energy is small, already represented by the chosen static density functional, or cancelled by a constraint in the relevant operator channel.

3 Many-body response, screening, and conservation laws

3.1 Why screening, polarization, and the vertex are not independent

In Hedin's closed set of equations, the irreducible polarization P, dielectric function ϵ, screened interaction W, and reducible density response χ satisfy schematically [Hedin, 1965, Giustino, 2017]

$$P = -\mathrm{i}GG\Gamma_\rho, \tag{10}$$

$$\epsilon = 1 - vP, \tag{11}$$

$$W = \epsilon^{-1}v, \tag{12}$$

$$\chi = P(1 - vP)^{-1}. \tag{13}$$

Here $\Gamma_\rho = -\delta G^{-1}/\delta V_{\mathrm{tot}}$ is a proper density vertex in the stated convention. Matrix products and integrations over internal variables are implicit. These relations answer an important conceptual question: screening, the quasiparticle vertex, and static density response are not three independent physical factors available for arbitrary multiplication. The vertex enters P; P determines ϵ; and χ follows from both. Mixing proper and screened vertex conventions can therefore count screening twice.

In a scalar electron gas, a convenient schematic form for the quasiparticle coupling to an external ionic perturbation is

$$g^{\mathrm{phys}}(\boldsymbol{q}, \omega) = g^0(\boldsymbol{q})\,\frac{z\,\Gamma_\rho(\boldsymbol{q}, \omega)}{\epsilon(\boldsymbol{q}, \omega)}, \tag{14}$$

where z is the quasiparticle residue. This equation is useful only after the vertex convention and screening factor have been stated. In this scalar expression $z\Gamma_\rho$ includes the quasiparticle external-leg residue appropriate to that convention. It must not be identified without conversion with the fixed-basis inverse-Green-function vertex of equation (9).

3.2 The Ward identity and its two limits

Charge conservation and gauge invariance imply the Ward–Takahashi identity [Nambu, 1960]

$$\mathrm{i}\nu\,\Gamma^0(k+q, k) - \boldsymbol{q}\cdot\boldsymbol{\Gamma}^J(k+q, k) = G^{-1}(k+q) - G^{-1}(k), \tag{15}$$

up to sign conventions for the Fourier transform and electric charge. Γ^0 is the scalar charge vertex and $\boldsymbol{\Gamma}^J$ is the current vertex. The identity constrains the full four-vector vertex; the conclusion depends on how $(\boldsymbol{q}, \nu)$ approaches zero.

If one sets $\boldsymbol{q} = 0$ first and then takes $\nu \to 0$, the dynamic uniform limit is

$$\Gamma^0_\omega = \frac{\partial G^{-1}}{\partial(\mathrm{i}\omega)} = 1 - \frac{\partial \Sigma}{\partial(\mathrm{i}\omega)} = z^{-1}, \qquad z\Gamma^0_\omega = 1. \tag{16}$$

This is an exact cancellation for the conserved total charge in that limit.

If instead one sets $\nu = 0$ first and then lets $\boldsymbol{q} \to 0$, the Ward identity directly constrains the longitudinal current vertex. The static scalar response also contains the response of the self-energy to chemical potential and the Landau interaction. For an isotropic Fermi liquid, for example,

$$\frac{\partial n}{\partial \mu} = \frac{N^*(0)}{1 + F_0^s}, \tag{17}$$

not simply $zN_0(0)$. Therefore $z\Gamma_\rho = 1$ is not a universal static density-vertex identity.

For an acoustic phonon, $\omega_{\nu\boldsymbol{q}} \simeq c_s|\boldsymbol{q}|$ and generally $v_F/c_s \gg 1$, so

$$\frac{v_F|\boldsymbol{q}|}{\omega_{\nu\boldsymbol{q}}} \simeq \frac{v_F}{c_s} \gg 1. \tag{18}$$

This is the adiabatic or static side of the response. The dynamic result in equation (16) cannot by itself explain the success of ordinary DFPT for acoustic phonons.

3.3 The exact common-potential anchor

There is nevertheless a clear physical picture in the strict uniform case. If a lattice displacement produces a potential $H' = Du\hat{N}$ that couples only to the conserved total particle number, every many-electron state with N electrons shifts by NDu. The energy required to add one electron therefore shifts by exactly Du, irrespective of the strength of interactions inside the system.

Equivalently, a common scalar shift can be absorbed into the frequency origin,

$$\Sigma_u(\omega) = \Sigma_0(\omega - Du), \qquad \partial_u\Sigma = -D\,\partial_\omega\Sigma. \tag{19}$$

The shift of the quasiparticle pole is then

$$z\big[D + \partial_u\Sigma\big] = D. \tag{20}$$

This derivation explains the cancellation physically: an interaction cannot renormalize a change of the energy zero. It is an exact anchor at strict $\boldsymbol{q} = 0$ for a common charge potential, not a proof at finite momentum and not a statement about orbital splitting, bond modulation, or other internal operators.

What conservation does for this project. Conservation laws identify one protected full-system direction: a strict uniform $\boldsymbol{q} = 0$ scalar potential coupled to total charge. A projected `global_charge` component may use this anchor only after the original unprojected perturbation is verified to be that full-space common shift. Relative site/block identities (`site_charge`), block-internal traceless operators (`internal`), and inter-block operators (`nonlocal`) are not protected by this argument.

---

4 What the uniform electron gas does and does not explain

4.1 Uniform equilibrium does not mean only zero momentum

The UEG has a constant equilibrium density, but it can be perturbed at every wavevector. A scalar external potential has the form

$$\delta V(\boldsymbol{r}) = \delta V_{\boldsymbol{q}} e^{i\boldsymbol{q}\cdot\boldsymbol{r}}, \qquad \hat{\rho}_{\boldsymbol{q}} = \sum_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}+\boldsymbol{q}} c_{\boldsymbol{k}}. \tag{21}$$

Translational invariance makes different wavevectors independent, $\chi(\boldsymbol{q}, \boldsymbol{q}') = \delta_{\boldsymbol{q}\boldsymbol{q}'}\chi(\boldsymbol{q})$; it does not eliminate finite $\boldsymbol{q}$. Only $\hat{\rho}_{\boldsymbol{0}} = \hat{N}$ is conserved. At finite $\boldsymbol{q}$ the perturbation is a density wave, and no exact identity says its quasiparticle coupling is unrenormalized.

For elastic scattering on a spherical Fermi surface,

$$q = 2k_F \sin(\theta/2), \tag{22}$$

so physically relevant scattering spans $0 \leq q \leq 2k_F$. A theory that works only at $q = 0$ does not yet predict transport or pairing.

4.2 The quantitative DFPT–many-body matching condition

Using the scalar relations of sections 2 and 3, denote the many-body irreducible polarization by P_{MB} and match it to a static local kernel,

$$P_{\mathrm{MB}}(q, 0) = \frac{\chi_s(q, 0)}{1 - f_{\mathrm{xc}}(q, 0)\chi_s(q, 0)}. \tag{23}$$

Then

$$1 - (v + f_{\mathrm{xc}})\chi_s = (1 - vP_{\mathrm{MB}})(1 - f_{\mathrm{xc}}\chi_s). \tag{24}$$

The correction applied to an already screened DFPT vertex must be the complete physical-to-DFPT ratio

$$K_{\mathrm{total}}(q, 0) \equiv \frac{g^{\mathrm{phys}}(q, 0)}{g^{\mathrm{DFPT}}(q, 0)} = z\Gamma_\rho(q, 0)\frac{1 - [v(q) + f_{\mathrm{xc}}(q, 0)]\chi_s(q, 0)}{1 - v(q)P_{\mathrm{MB}}(q, 0)}. \tag{25}$$

Under the scalar matching above and the corresponding proper-vertex relation

$$z\Gamma_\rho(q, 0) \simeq \frac{P_{\mathrm{MB}}(q, 0)}{\chi_s(q, 0)}, \tag{26}$$

the factors in equation (25) cancel and $K_{\mathrm{total}} \simeq 1$. Thus P_{MB}/χ_s is a relation for the proper vertex in this convention, not a second correction factor to multiply onto the already screened DFPT perturbation. The matching is not produced by the dynamic Ward identity alone.

Recent vertex-diagrammatic Monte Carlo calculations for the electron liquid at $r_s = 1$–5 report that the fully interacting quasiparticle vertex and a DFPT-like expression with a static exchange–correlation kernel agree accurately over momenta extending to $2k_F$ [Cai et al., 2025]. The residual dependence on the incoming fermion momentum is weak, with backscattering providing the most demanding comparison. This is nontrivial numerical evidence for equation (25) across finite momentum; it is not an exact theorem.

4.3 Why the UEG is exceptionally simple

The UEG contains a scalar density channel but no atomic orbitals, sublattices, crystal-field splittings, bond angles, hybridization matrix, Umklapp mixing, or reciprocal-space local-field matrix. Its low-energy perturbation space is therefore effectively one-dimensional. Moreover, local and semilocal density functionals are built from electron-gas information, making the UEG the most favorable setting for a density-kernel closure.

This explains both the value and the limitation of the UEG result:

- it provides evidence that the complete static DFPT vertex can be accurate in the tested scalar-density domain, including finite q;

- it cannot determine the correction to orbital, bond, hopping, spin, or separate two-body interaction vertices that do not exist in the model; and
- its finite-q agreement must be tested, not assumed, when transferred to a crystal with local-field and multi-orbital structure.

UEG conclusion. The UEG does not show that real materials have only $q = 0$ coupling. It shows that when the perturbation space contains only density, the complete screened DFPT vertex can be accurate over a surprisingly wide momentum range. It does not license a further P_{MB}/χ_s multiplication. Whether an analogous accuracy or compressible correction exists in solid-state one-body channels is a separate working hypothesis.

5 From a common shift to real material perturbations

5.1 Projection and operator decomposition

Let P_c project onto a localized low-energy or correlated subspace, such as transition-metal d orbitals and ligand p orbitals. The projected DFPT perturbation is

$$D_{\nu\boldsymbol{q}} = P_c\, \partial_{\nu\boldsymbol{q}} V_{\mathrm{KS}}\, P_c. \tag{27}$$

For the current executable decomposition, $D_{\nu\boldsymbol{q}}$ denotes a supported Hermitian representation: a real-space derivative, Gamma perturbation, or real $\boldsymbol{q}/-\boldsymbol{q}$ standing wave. A general fixed-$\boldsymbol{q}$ component instead satisfies equation (3) and requires a future paired interface.

Let site block i have dimension n_i, let $N = \sum_i n_i$, and define $\bar{d} = \operatorname{Tr} D/N$ and $d_i = \operatorname{Tr} D_{ii}/n_i$. The four one-body components are

$$\begin{aligned} D^{\mathrm{global\_charge}} &= \bar{d}\, I_N, && (28)\\ D^{\mathrm{site\_charge}}_{ii} &= (d_i - \bar{d}) I_i, && (29)\\ D^{\mathrm{internal}}_{ii} &= D_{ii} - d_i I_i, && (30)\\ D^{\mathrm{nonlocal}}_{ij} &= D_{ij} \quad (i \neq j), && (31) \end{aligned}$$

with all unspecified blocks zero. Thus

$$D_{\nu\boldsymbol{q}} = D^{\mathrm{global\_charge}}_{\nu\boldsymbol{q}} + D^{\mathrm{site\_charge}}_{\nu\boldsymbol{q}} + D^{\mathrm{internal}}_{\nu\boldsymbol{q}} + D^{\mathrm{nonlocal}}_{\nu\boldsymbol{q}}. \tag{32}$$

The four components are mutually orthogonal under $\langle A, B\rangle_{\mathrm{HS}} = \mathrm{Tr}(A^\dagger B)$. Here `global_charge` is first an algebraic identity inside the projected space. It is Ward protected only if the original unprojected perturbation is separately verified to be a strict uniform $\boldsymbol{q} = 0$ common scalar potential over the full electronic Hilbert space. The condition $P_c D P_c \propto I_N$ alone is insufficient because complementary-space and cross-space blocks may differ. Moreover, adding a common cI changes the reported channel weight, so the energy-zero and chemical-potential convention must be fixed or that gauge shift must be removed before comparing weights. A relative block shift is `site_charge` and is unprotected. The decomposition is basis-dependent, so the project must state the Wannier or projector construction and test robustness against a reasonable basis change.

Derivatives such as $\partial U/\partial u$ and $\partial J/\partial u$ do not belong in equation (32). They multiply two-body operators, for example

$$\delta H^{(2)}_u = \frac{\partial U}{\partial u}\, \hat{n}_\uparrow \hat{n}_\downarrow + \frac{\partial J}{\partial u}\, \hat{O}_J + \cdots, \tag{33}$$

and require a separate two-body operator basis and response model. The present software does not implement that space.

5.2 Why a local common operator does not shift all bands equally

Suppose a phonon produces the same local potential D on all orbitals of the correlated subspace. A Bloch state still shifts by

$$\delta\varepsilon_{n\boldsymbol{k}} = D\langle\psi_{n\boldsymbol{k}}|P_c|\psi_{n\boldsymbol{k}}\rangle, \tag{34}$$

which depends on its correlated-orbital weight. At finite momentum,

$$g_{mn}(\boldsymbol{k},\boldsymbol{q}) = D\langle\psi_{m,\boldsymbol{k}+\boldsymbol{q}}|P_c|\psi_{n\boldsymbol{k}}\rangle, \tag{35}$$

which also depends on phases and interband overlaps. DFPT already computes these geometric and hybridization effects. The proposed correction concerns the many-body renormalization of the operator, not a replacement of all matrix elements by a common number.

5.3 Evidence from correlated materials

DFT+DMFT calculations provide a useful controlled contrast because they evaluate the derivative of a local, frequency-dependent correlated self-energy [Abramovitch et al., 2025]. For $SrVO_3$ with $U = 4.5$ eV and $J = 0.675$ eV, the reported zero-frequency coupling for an M-point Jahn–Teller mode increases from about 44 meV in DFPT to 87 meV in DFT+DMFT. The corresponding R-point breathing-mode coupling changes from about 58 meV to 50 meV. The same correlated material therefore contains a strongly corrected orbital-splitting mode and a comparatively stable block-identity density modulation. The finite-R classification must be made in the paper's $2 \times 2 \times 2$ real-displacement supercell, equivalently a real $\boldsymbol{q}/-\boldsymbol{q}$ standing wave: the V-site identity blocks alternate in phase and therefore form a `site_charge`-like pattern. A single primitive-cell $\mathrm{diag}(g, g, g)$ block would instead look algebraically like `global_charge` and is not a valid classification of the R-point mode. In either representation, finite momentum excludes the Ward protected direction.

For hole-doped $CaCuO_2$ at a representative interaction $U = 3.1$ eV, the reported zero-frequency half- and full-breathing couplings change from approximately 70 to 76 meV and from 53 to 45 meV, respectively [Abramovitch et al., 2025]. All Abramovitch numbers quoted here are comparisons within that paper's declared fixed-basis and many-body setup, not independent cross-code accuracy benchmarks. These static changes are modest even though the quasiparticle weight is reduced. At larger $U = 4.7$ eV, however, the same study finds strong vertex frequency dependence on the phonon energy scale. Agreement at $\omega = 0$ is then insufficient to validate a dynamic scattering calculation.

These cases support a mode-resolved interpretation, but they do not yet prove the specific decomposition in equation (32). Two further methods establish that standard DFPT has genuine, computable limitations:

- DFPT+U corrects both the reference electronic structure and the response of Hubbard occupations. In CoO, this changes the polar Fröhlich interaction relative to ordinary DFPT [Zhou et al., 2021].
- GW perturbation theory differentiates a nonlocal, energy-dependent GW self-energy. It produces correlation-enhanced electron–phonon interactions in $Ba_{1-x}K_xBiO_3$ [Li et al., 2019].

These are not competing anecdotes. They identify distinct missing channels: local correlated occupations in DFPT+U, local dynamic self-energy in DFT+DMFT, and nonlocal quasiparticle self-energy in GW perturbation theory. Earlier correlated-electron analyses likewise show that vertex corrections can be strongly momentum- and channel-dependent [Grilli and Castellani, 1994].

5.4 What "DFPT works in a strongly correlated material" can mean

A material may be strongly correlated according to its spectral weight, Hubbard bands, or mass enhancement while a particular phonon derivative $\partial_u \Sigma^{\mathrm{corr}}$ remains small or nearly common within the relevant low-energy subspace. Conversely, a seemingly small traceless orbital component can be strongly amplified near an orbital instability. The correction is controlled by the product of perturbation weight and channel susceptibility, not by either factor alone.

Accordingly, a successful DFPT result in one mode should not be generalized to all modes of that material. The correct unit of comparison is a material–phonon–momentum–frequency tuple.

6 Working physical hypothesis

The preceding theory suggests a hypothesis that is stronger than a statement of numerical cancellation and weaker than a claimed universal theorem.

Channel-selective protection hypothesis. For a low-energy electron–phonon process, the leading error of static DFPT is controlled by (i) the decomposition of the self-consistent phonon perturbation into the one-body `global_charge`, `site_charge`, `internal`, and `nonlocal` operators; (ii) any separately declared two-body interaction vertex; (iii) the many-body amplification in each operator space; and (iv) the momentum and frequency path of the response. Strong correlation changes the error mainly when the phonon has appreciable weight in an unprotected channel with a large or dynamic susceptibility.

Write the projected perturbation in an orthonormal operator basis,

$$D^{\mathrm{KS}}_{\nu\boldsymbol{q}} = \sum_a d_{a,\nu\boldsymbol{q}} O_a, \qquad \mathrm{Tr}(O_a^\dagger O_b) = \delta_{ab}. \tag{36}$$

The corresponding fixed-basis one-body inverse-Green-function vertex is represented by a channel kernel

$$\Lambda^{(1)}_{\nu\boldsymbol{q}}(\omega) = \sum_{ab} \mathcal{K}_{ab}(\boldsymbol{q},\omega) d_{b,\nu\boldsymbol{q}} O_a. \tag{37}$$

The first implementation may retain only diagonal or small-block kernels, but the general expression allows channel mixing. The hypothesis predicts the following.

1. `global_charge` has an exact consistency constraint only when the original unprojected operator is verified as a strict uniform $\boldsymbol{q} = 0$ full-space common shift: it is an energy-zero change and cannot acquire an arbitrary quasiparticle renormalization.
2. `site_charge` is not protected. Finite-q UEG evidence instead says that the complete screened DFPT vertex is accurate in the tested scalar-density domain; transfer to crystals remains a hypothesis.
3. `internal` and `nonlocal` one-body operators, and separate two-body interaction derivatives, are not constrained by charge conservation and may receive large, mode-specific corrections.
4. Static and dynamic validity are separate. A channel may have $\mathcal{K}_a(\boldsymbol{q}, 0) \simeq 1$ while varying rapidly across $\omega_{\nu\boldsymbol{q}}$.

6.1 Measurable channel weights

A simple Frobenius norm of an operator can overemphasize matrix elements that do not connect the relevant electronic states. For a target energy window $\mathcal{W}$ and nonnegative sampling weights $w_{n\boldsymbol{k}}$, define a Fermi-window channel strength

$$S_{a,\nu\boldsymbol{q}} = \sum_{mn\boldsymbol{k}\in\mathcal{W}} w_{n\boldsymbol{k}} w_{m,\boldsymbol{k}+\boldsymbol{q}} \left|\langle\psi_{m,\boldsymbol{k}+\boldsymbol{q}}|d_{a,\nu\boldsymbol{q}}O_a|\psi_{n\boldsymbol{k}}\rangle\right|^2. \tag{38}$$

Normalize $w_a = S_a/\sum_b S_b$ for interpretation. These weights answer which operators the actual low-energy scattering process samples, not merely which matrix block is largest in a localized representation. The current executable weights apply only to the four supported Hermitian one-body components; they do not include the two-body space of equation (33).

A correction-risk score should combine channel weight and amplification. A minimal form is

$$R_{\nu\boldsymbol{q}} = \left[\sum_a w_{a,\nu\boldsymbol{q}} \left|\mathcal{K}_a(\boldsymbol{q},0)-1\right|^2\right]^{1/2} + \lambda_{\mathrm{dyn}} \max_a \eta^{\mathrm{dyn}}_{a,\nu\boldsymbol{q}}, \tag{39}$$

where

$$\eta^{\mathrm{dyn}}_{a,\nu\boldsymbol{q}} = \omega_{\nu\boldsymbol{q}} \left|\partial_\omega \ln \mathcal{K}_a(\boldsymbol{q},\omega)\right|_{\omega=0}. \tag{40}$$

The threshold and λ_{dyn} are calibration parameters, not universal constants.

6.2 Falsification criteria

The hypothesis must be rejected or revised if any of the following occur across a held-out benchmark set:

- channel weights do not correlate with the size or sign of the beyond-DFPT correction;
- nominally `site_charge`-dominated or other nonuniform density modes show large, uncontrolled corrections that the fitted channel response cannot explain;
- the kernel requires essentially independent parameters for every band–momentum matrix element, eliminating the proposed compression; or
- dynamic effects dominate so broadly that a static risk gate cannot identify a useful low-cost regime.

The theory will then need an enlarged operator basis, nonlocal channel mixing, or a fully frequency-dependent formulation. A negative result would still identify why the proposed simplification fails.

7 A next-generation corrected-DFPT framework

The proposed method is a correction layer, not a replacement for the mature machinery of DFPT. It uses DFPT to obtain phonons, the fully self-consistent Kohn–Sham perturbation, and its Bloch-state geometry, then spends many-body effort only on a low-dimensional projected operator.

7.1 Corrected perturbation and matrix element

In a diagonal-channel approximation, define the fixed-basis one-body vertex

$$\Lambda^{(1),\mathrm{next}}_{\nu\boldsymbol{q}}(\omega) = D^{\mathrm{KS}}_{\nu\boldsymbol{q}} + \sum_a \left[\mathcal{K}_a(\boldsymbol{q},\omega)-1\right] D^{(a)}_{\nu\boldsymbol{q}}. \tag{41}$$

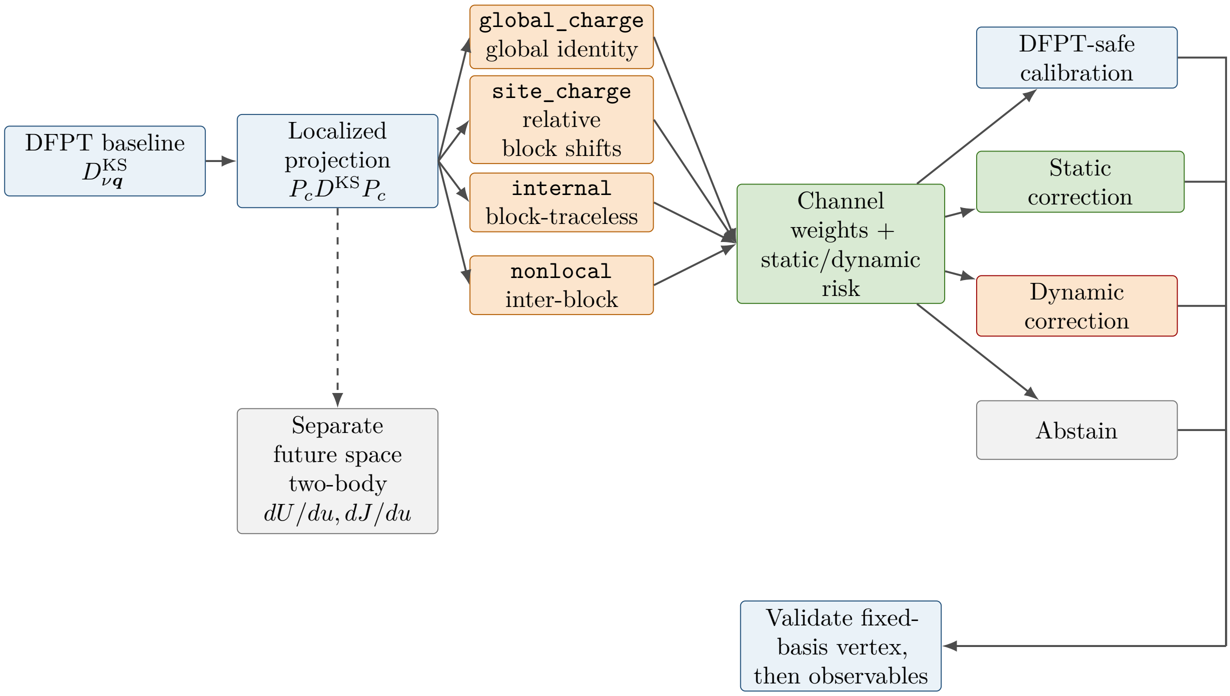

Figure 1: Proposed channel-resolved workflow. DFPT supplies the baseline perturbation. Projection and operator decomposition expose which physical one-body channels a phonon drives and records any separate two-body vertex. A risk gate returns one of four outputs: DFPT-safe calibration candidate, static correction, dynamic correction, or abstention. Validation is performed first on a convention-fixed vertex and only then on observables.

Its matrix element in the declared reference basis is

$$g^{\mathrm{vertex}}_{mn\nu}(\boldsymbol{k},\boldsymbol{q},\omega) = \langle \psi_{m,\boldsymbol{k}+\boldsymbol{q}} | \Lambda^{(1),\mathrm{next}}_{\nu\boldsymbol{q}}(\omega) | \psi_{n,\boldsymbol{k}} \rangle. \tag{42}$$

When all $\mathcal{K}_a = 1$, the method reduces exactly to its DFPT baseline. Because the correction acts on operators before Bloch-state matrix elements are taken, it preserves the band, momentum, and phase dependence already present in DFPT.

For isolated quasiparticle poles with band-diagonal scalar residues, a physical quasiparticle matrix element also requires the external-leg residues and a rotation to quasiparticle states:

$$g^{\mathrm{QP}}_{mn} = \langle \phi^{\mathrm{QP}}_m | Z^{1/2}_m \Lambda^{(1),\mathrm{next}} Z^{1/2}_n | \phi^{\mathrm{QP}}_n \rangle. \tag{43}$$

Those transformations are not part of the fitted $\mathcal{K}$ and must not be silently included in only some training anchors. Degenerate or strongly mixed multiorbital poles generally require matrix residues and properly normalized left and right pole eigenvectors rather than the scalar form in equation (43).

The executable quantitative prototype generalizes the diagonal rescaling in a fixed operator basis. Let O_a be localized, symmetry-adapted operators and write the short-range perturbation as

$$D^{\mathrm{SR}}_s = \sum_a c^{\mathrm{DFPT}}_{sa} O_a, \tag{44}$$

where s labels a material–mode–momentum–frequency sample and c^{DFPT}_{sa} is its DFPT coefficient. The four one-body channel names are `global_charge`, `site_charge`, `internal`, and `nonlocal`. For convention-matched coefficients of the fixed-basis one-body inverse-Green-function vertex, the static sparse-anchor hypothesis is

$$c_{sa}^{\mathrm{ref}} = \sum_b K_{ab} c_{sb}^{\mathrm{DFPT}} + r_{sa}. \tag{45}$$

Here K_{ab} is a small response matrix and r_{sa} is the residual. Diagonal elements rescale one channel; off-diagonal elements represent channel mixing. $K = I$ recovers the DFPT coefficient vector. Equation (45) is a falsifiable low-dimensional hypothesis, not a statement that the complete band- and momentum-resolved electron–phonon matrix is low rank.

The current standard-library implementation fits K from supplied coefficient vectors, predicts held-out vectors, and reports their error. It does not yet construct the O_a from a production DFPT or Wannier output. A real application must declare the basis and gauge, keep all phonon and matrix-element normalization conventions fixed, and separate analytic long-range polar fields from the localized short-range operator before fitting. The bundled synthetic response example excites only `global_charge`, `internal`, and `nonlocal`; its omitted `site_charge` coefficient is zero by construction, so it is not a four-channel physical validation.

7.2 How each channel is corrected

`global_charge` and `site_charge`. The implementation enforces the exact common-potential constraint only for `global_charge` with explicit evidence that the original unprojected perturbation is a full-space uniform shift at $\boldsymbol{q} = 0$. `site_charge` contains relative block-identity shifts and is a correction-risk channel. At finite q, the UEG comparison constrains the complete ratio K_{total} in equation (25), which is approximately one in the cited calibration domain. It does not provide P_{MB}/χ_s as an extra kernel for an already screened DFPT input. Crystal local-field matrices, dimensionality, polar long-range fields, and multi-band screening require their own convention-matched tests.

Local internal channels. For orbital splitting or mixing in a correlated subspace, obtain $\partial_u \Sigma_{ab}(\omega)$ from finite-displacement DFT+DMFT, a linearized impurity response, or a validated low-energy model. If $\eta_a^{\mathrm{dyn}} \ll 1$, retain only $\mathcal{K}_a(\boldsymbol{q}, 0)$. Near orbital, spin-state, or Mott instabilities, the dynamic kernel and off-diagonal channel mixing must be kept.

Nonlocal channels. When a phonon mainly changes hopping, hybridization, or long-range exchange, a local impurity self-energy is insufficient. Targeted GW perturbation theory, finite-difference quasiparticle calculations, or a nonlocal embedding method can provide selected matrix elements of $\mathcal{K}_{ab}$ [Li et al., 2019]. Symmetry and locality can then interpolate those corrections over equivalent bonds and wavevectors.

Separate two-body interaction vertices. If a displacement changes U, J, or another downfolded interaction, compute those derivatives with constrained response and include the resulting vertex in a separate two-body operator space. Do not add it to the one-body Hilbert–Schmidt decomposition or fit it with the present K_{ab}. Constrained DFPT provides a precedent for separating screening processes during downfolding [Nomura and Arita, 2015].

7.3 Four-output decision gate

The framework chooses the least expensive level justified by diagnostics.

Table 1: Decision levels for the proposed method.

Level	Physical diagnostic	Calculation	Required check
DFPT-safe	`global_charge` dominated; original operator explicitly verified as a strict uniform $\boldsymbol{q} = 0$ full-space common shift; adequate reference; weak estimated dynamics	Standard DFPT matrix elements	Gauge anchor and one representative high-level control
Static correction	`site_charge`, `internal`, or `nonlocal` weight is appreciable, but η^{dyn} is small	Channel-projected static $\mathcal{K}_a(\boldsymbol{q}, 0)$	Held-out modes and momenta
Dynamic correction	Rapid vertex variation, soft electronic mode, small coherence scale, or proximity to an instability	Frequency-dependent $\mathcal{K}_{ab}(\boldsymbol{q}, \omega)$	Values across the physical phonon-frequency window
Abstain	Outside the validated channel/basis/material domain	No automatic prediction	Request a higher-level calculation

7.4 Uncertainty and invariance requirements

Every reported g^{next} should include uncertainty from high-level training data, operator-basis choice, momentum interpolation, and neglected channel mixing. The following tests are mandatory:

- invariance to unitary rotations within a declared degenerate orbital subspace;
- the acoustic sum rule and appropriate translational constraints;
- the uniform scalar-potential gauge test;
- Hermiticity for the supported real-space/Gamma/standing-wave input and $D(\boldsymbol{q})^\dagger = D(-\boldsymbol{q})$ for a future paired finite-$\boldsymbol{q}$ interface; and
- recovery of uncorrected DFPT with $\mathcal{K} = I$.

7.5 Why this is a new prediction framework

Existing beyond-DFPT methods demonstrate how to compute particular corrections, but they remain expensive when applied to all modes and wavevectors. The proposed novelty is the combination of (i) an operator-level physical diagnosis, (ii) a four-output choice of correction theory, (iii) a candidate low-dimensional kernel, and (iv) explicit uncertainty and abstention. Its success criterion is not that it matches a full many-body calculation by construction, but that it predicts which parts of the DFPT matrix element need correction and reaches a defined accuracy with far fewer high-level calculations.

This correction layer is not cheaper than performing the one DFPT calculation that supplies its input. Its immediate cost advantage, if validated, is relative to evaluating a beyond-DFPT vertex for every mode and momentum. Beating DFPT itself would require an additional surrogate that predicts the DFPT channel coefficients without evaluating DFPT at every requested point. That surrogate is neither implied by the channel decomposition nor implemented in the present software.

8 AI-agent research and implementation program

The AI agent is not used to replace physical validation. Its role is to maintain traceability across literature, material cases, calculations, code, and failed hypotheses, while automating the repetitive parts of comparison and model construction.

8.1 Closed-loop workflow

For each candidate material and phonon mode, the agent executes the following auditable loop.

1. **Read and extract.** Search primary literature and record the material, mode eigenvector, $\boldsymbol{q}$, electronic method, interaction parameters, definition and normalization of g, frequency at which the vertex is evaluated, and reported uncertainty.
2. **Classify the perturbation.** Project the DFPT perturbation, compute the `global_charge`, `site_charge`, `internal`, and `nonlocal` one-body weights of equation (38), and attach a human-readable physical interpretation. Record interaction derivatives in a separate two-body space.
3. **Select a benchmark.** Choose the least expensive many-body method able to represent the active object: UEG/vertex Monte Carlo for the complete scalar-density comparison, DFT+DMFT for local correlated channels, GW perturbation theory for nonlocal quasiparticle response, or an explicit model when interaction derivatives are central.
4. **Generate a hypothesis.** Propose a low-dimensional form of $\mathcal{K}_{ab}(\boldsymbol{q}, \omega)$ with symmetry, Ward/gauge, locality, and causality constraints encoded before fitting.
5. **Design a discriminating calculation.** Select the mode or momentum point that maximizes disagreement between the current hypothesis and a plausible alternative, instead of merely adding more similar data.
6. **Implement and test.** Produce a versioned correction module, unit tests for algebraic invariants, and numerical regression tests against held-out high-level vertices.
7. **Update or reject.** Store negative evidence and revise the claim ledger. The agent may not promote an empirical relation to an exact principle.

Each extracted number must retain source, equation/table/figure location, normalization convention, and whether it was read directly or inferred. This is essential because electron–phonon matrix elements can differ by phonon normalization, spin factors, mode phase, orbital gauge, and screening convention.

8.2 Minimal benchmark ladder

The proposed ladder is designed to distinguish physical channels, not simply to cover many chemical compositions.

The UEG is the first diagnostic object because it isolates density physics. It must not be the only validation object, because success there cannot test internal crystal channels. Continuous fixed-q UEG comparison is presently a literature/theory calibration; the Hermitian executable does not claim that interface until a paired $q/-q$ or real-space implementation exists.

Table 2: Minimal benchmark ladder for developing and falsifying the method.

System	Active contrast	Reference level	Question answered
Uniform electron gas	$q = 0$ anchor versus $0 < q \leq 2k_F$ density response	Vertex diagrammatic Monte Carlo	Is the finite-q density prior accurate and what controls its residual?
$SrVO_3$	Jahn–Teller versus breathing modes in a phase-correct real supercell or standing wave	DFT+DMFT	Does traceless orbital weight predict the large mode-selective correction?
Doped $CaCuO_2$	Half- versus full-breathing, static versus dynamic vertex	DFT+DMFT	Can the dynamic-risk indicator detect failure of a static correction?
CoO	Correlated insulator and polar coupling	DFPT+U / experiment	Does the framework detect an invalid reference state and occupation-response channel?
$Ba_{1-x}K_xBiO_3$ or a simpler GWPT solid	Nonlocal quasiparticle response	GW perturbation theory	Can a sparse nonlocal kernel reproduce the high-level correction?

8.3 Accuracy, speed, and transparency metrics

The target should be evaluated against three independent axes.

Accuracy. Report the error of complex matrix elements where phases are comparable, squared couplings averaged on the relevant energy shell, and final observables. A real-material benchmark must predeclare separate acceptance bounds for the projected operator, the gauge-invariant coupling measure, and the target observable. The present synthetic held-out example cannot determine those bounds and is not counted as a physical-accuracy result.

Cost. Let N_{full} be the number of mode–momentum points in each campaign, N_{hi} the number of convention-matched higher-level calibration anchors used once to fit the response matrix, and N_{camp} the number of campaigns that reuse that fit. With per-point costs c_{DFPT}, c_{hi}, and c_{inf}, and one-time training cost C_{train}, the accounting model is

$$C_{\text{DFPT}} = N_{\text{camp}} N_{\text{full}} c_{\text{DFPT}}, \tag{46}$$

$$C_{\text{dense-hi}} = N_{\text{camp}} N_{\text{full}} \left(c_{\text{DFPT}} + c_{\text{hi}}\right), \tag{47}$$

$$C_{\text{corrected}} = C_{\text{train}} + N_{\text{hi}} c_{\text{hi}} + N_{\text{camp}} N_{\text{full}} \left(c_{\text{DFPT}} + c_{\text{inf}}\right). \tag{48}$$

The current correction model still requires DFPT coefficients at every predicted point, so it cannot be cheaper than DFPT alone: $C_{\text{corrected}} \geq C_{\text{DFPT}}$. It is useful only when the alternative is a higher-level correction at every point and $C_{\text{corrected}} < C_{\text{dense-hi}}$. The bundled software contract uses normalized declared costs and gives $C_{\text{DFPT}} = 100$, $C_{\text{dense-hi}} = 600$, and $C_{\text{corrected}} = 122$; the dense higher-level baseline therefore costs 4.918 times as much as the corrected path, while the corrected path has a cost overhead of 22 relative to DFPT alone. This verifies the accounting code, not a measured runtime. Beating DFPT itself would require a separate surrogate for the DFPT coefficients. A publishable speed claim must count actual core or GPU hours, memory, and wall time at matched accuracy and compare against a converged, symmetry-reduced DFPT plus Wannier/EPW interpolation workflow rather than a naive dense-point construction.

Transparency. For every prediction, output the operator decomposition, selected correction level, training-source distance, uncertainty, and any violated invariant. The framework should expose why a mode was corrected rather than only returning a number.

8.4 Implementation layers

The initial software can be separated into small testable modules:

1. a reader for DFPT perturbation matrices and wavefunctions;
2. localized-subspace construction and gauge checks;
3. operator decomposition and Fermi-window channel weights;
4. adapters for high-level reference data from UEG, DFT+DMFT, DFPT+U, and GWPT;
5. constrained kernel fitting/interpolation with uncertainty;
6. corrected-matrix-element reconstruction; and
7. validation reports separating matrix-element and observable gates.

No production implementation should begin before the two-day literature sprint freezes the first hypothesis, the exact comparison quantities, and the first two material-mode contrasts.

The repository implementation is intentionally one layer below such production use. It fits and applies equation (45), scores synthetic held-out coefficient vectors, and evaluates equation (48). The supplied exact synthetic result tests the software path only; real-material fitting, measured timing, uncertainty calibration, and transfer across modes remain open gates. The recorded unit-test/evaluation counts, synthetic RMSE, and normalized cost ratio are all software-contract outputs; none is a measured material vertex, physical accuracy result, or speedup.

9 Conclusions and immediate decision

The apparent broad usefulness of DFPT should not be described as a mysterious global cancellation. DFPT computes a full, state-dependent static Kohn–Sham response. The missing many-body object is the phonon derivative of the beyond-Kohn–Sham self-energy. That derivative can be small for one perturbation and large for another even when the equilibrium self-energy and mass enhancement are large.

The clearest physical anchor is an unprojected, full-space uniform common potential coupled to conserved total charge: it only changes the energy zero, so interactions cannot arbitrarily renormalize it. A projected identity component alone is not sufficient. The Ward identity states this constraint precisely, but its dynamic and static limits must not be confused. The UEG extends the picture with strong numerical evidence that a static density kernel remains accurate over finite momenta up to $2k_F$ in the tested regime. Because the UEG contains only the density channel, this result suggests a compression principle rather than a universal proof.

Real crystals add relative `site_charge`, block-internal `internal`, inter-block `nonlocal`, spin, and separate two-body interaction-modulation operators. Existing DFT+DMFT, DFPT+U, and GWPT results show that their corrections can be strongly mode-selective. This motivates the proposal's central hypothesis: the useful small parameter is not "correlation strength" by itself, but the weight of a phonon in each operator channel times the many-body amplification and dynamical variation of that channel.

The proposed next-generation calculation therefore retains DFPT, decomposes its perturbation, and applies a channel kernel only where diagnostics demand it. This can be simpler and cheaper than computing a complete many-body vertex at every mode and momentum while being more physically transparent than a black-box correction to every matrix element. It is not faster than the single DFPT calculation used as its baseline. Its current computational claim is narrower: sparse higher-level anchors plus cheap inference can cost less than evaluating a higher-level correction at every point. A future speed claim against DFPT would require a separately implemented surrogate for the DFPT coefficients and a matched comparison with modern DFPT interpolation.

The accompanying software now fits a small channel-response matrix, predicts held-out coefficient vectors, reports error, and compares DFPT-only, dense higher-level, and sparsely corrected cost baselines. It explicitly reports that the present corrected path is not faster than DFPT. Its bundled held-out vectors are synthetic and its costs are normalized assumptions. These results establish reproducible software behavior, not material accuracy or a measured runtime speedup.

The executable one-body decomposition is presently Hermitian and therefore limited to real-space derivatives, Gamma, or real $q/-q$ standing waves. General fixed-q pairs and a separate two-body interaction-vertex space remain future interfaces. The fitted target is a fixed-basis inverse-Green-function vertex; external-leg $Z^{1/2}$ factors and quasiparticle-state rotations are separate.

First decision after submission. The first research milestone should test one sharp statement: in $SrVO_3$, does a basis-robust measure of `internal` weight separate the strongly enhanced Jahn–Teller mode from the weakly corrected finite-q `site_charge`-like breathing mode when both are represented in the phase-correct real supercell/standing-wave basis, while the complete UEG ratio supplies a scalar-density control? A positive result justifies building the static prototype; a negative result forces the hypothesis to be revised before software expansion.

A Detailed derivations and convention checks

A.1 DFPT response factorization in a scalar system

Starting from equations (5) and (6),

$$\delta V_{\mathrm{KS}} = \delta V_{\mathrm{ion}} + (v + f_{\mathrm{xc}})\chi_s \delta V_{\mathrm{KS}}, \tag{49}$$

$$\big[1 - (v + f_{\mathrm{xc}})\chi_s\big]\delta V_{\mathrm{KS}} = \delta V_{\mathrm{ion}}. \tag{50}$$

With $P_{\mathrm{MB}} = \chi_s/(1 - f_{\mathrm{xc}}\chi_s)$,

$$(1 - vP_{\mathrm{MB}})(1 - f_{\mathrm{xc}}\chi_s) = 1 - f_{\mathrm{xc}}\chi_s - v\frac{\chi_s}{1 - f_{\mathrm{xc}}\chi_s}(1 - f_{\mathrm{xc}}\chi_s) \tag{51}$$

$$= 1 - (v + f_{\mathrm{xc}})\chi_s. \tag{52}$$

Thus

$$g^{\mathrm{DFPT}} = \frac{g^0}{(1 - vP_{\mathrm{MB}})(1 - f_{\mathrm{xc}}\chi_s)}. \tag{53}$$

Comparing with $g^{\mathrm{phys}} = g^0 z\Gamma_\rho/(1 - vP_{\mathrm{MB}})$ gives the complete ratio

$$K_{\mathrm{total}} = z\Gamma_\rho \frac{1 - (v + f_{\mathrm{xc}})\chi_s}{1 - vP_{\mathrm{MB}}} = z\Gamma_\rho(1 - f_{\mathrm{xc}}\chi_s). \tag{54}$$

If the separately tested proper-vertex matching is $z\Gamma_\rho = P_{\mathrm{MB}}/\chi_s = (1 - f_{\mathrm{xc}}\chi_s)^{-1}$, then $K_{\mathrm{total}} \simeq 1$. The proper-vertex factor is therefore not an additional kernel to apply to g^{DFPT}.

A.2 Uniform-potential cancellation at a quasiparticle pole

Let the pole satisfy

$$E_u - \varepsilon - Du - \Sigma_0(E_u - Du) = 0. \tag{55}$$

Differentiate at $u = 0$:

$$\frac{dE}{du} - D - \partial_\omega\Sigma_0\left(\frac{dE}{du} - D\right) = 0. \tag{56}$$

Provided the pole is well-defined, $(1 - \partial_\omega\Sigma_0)(dE/du - D) = 0$, so $dE/du = D$. In the alternative form where $\partial_u\Sigma = -D\partial_\omega\Sigma$, the same result is $z(D + \partial_u\Sigma) = D$ with $z = (1 - \partial_\omega\Sigma)^{-1}$. The result follows from the common shift of frequency and chemical potential, not from weak interaction.

A.3 Limit ordering in the Ward identity

Expand the right-hand side of equation (15) for small four-momentum:

$$G^{-1}(k + q) - G^{-1}(k) = \mathrm{i}\nu\, \partial_{\mathrm{i}\omega}G^{-1} + \boldsymbol{q}\cdot\nabla_{\boldsymbol{k}}G^{-1} + \mathcal{O}(q^2). \tag{57}$$

At $\boldsymbol{q} = 0$, matching the coefficient of $\mathrm{i}\nu$ gives the scalar dynamic vertex. At $\nu = 0$, matching the coefficient of $\boldsymbol{q}$ gives the longitudinal current vertex. A static scalar perturbation also changes the distribution and self-consistent field, producing the compressibility vertex. This explicit expansion shows why swapping the two limits changes the conclusion.

A.4 Matrix generalization

In a crystal, v, P, χ_s, f_{xc}, and ϵ are matrices in reciprocal vectors $\boldsymbol{G}, \boldsymbol{G}'$. Products do not generally commute, so the scalar factorization in equation (24) cannot be transferred by simply replacing every scalar with a matrix. A crystal implementation must retain matrix ordering and local-field components, or derive a controlled projected version of the relation. This is one of the first theoretical tasks of the project.

B Evidence matrix

Table 3: Current evidence for and against the working hypothesis. Numerical values follow the normalization reported in the cited studies and must be checked again before cross-code fitting.

System / mode	DFPT or baseline	Higher-level result	Dominant interpretation	Status and use
Strict full-space uniform common potential	Bare/common shift D	Exact pole shift D	Conserved total charge / energy-zero shift	Exact anchor only after the original unprojected operator is verified; no finite-q scattering claim
UEG, $0 \leq q \leq 2k_F$, $r_s = 1$–5	Static DFPT-like density kernel	Vertex diagrammatic Monte Carlo	Single scalar density channel	Recent preprint evidence for finite-q compression; not a theorem [Cai et al., 2025]
$SrVO_3$, M-point Jahn–Teller	about 44 meV	about 87 meV at $\omega = 0$	Traceless orbital splitting	Strong positive example for an unprotected internal channel [Abramovitch et al., 2025]
$SrVO_3$, R-point breathing	about 58 meV	about 50 meV at $\omega = 0$	Finite-q `site_charge`-like modulation in the phase-correct $2 \times 2 \times 2$ real supercell	Same-paper fixed-setup control; modest static correction [Abramovitch et al., 2025]
Doped $CaCuO_2$, half breathing	about 70 meV	about 76 meV at $\omega = 0$	`site_charge` / `nonlocal` candidate	Same-paper fixed-setup control; modest static correction at representative U [Abramovitch et al., 2025]
Doped $CaCuO_2$, full breathing	about 53 meV	about 45 meV at $\omega = 0$	`site_charge` / `nonlocal` candidate	Same-paper fixed-setup control; modest negative static correction at representative U [Abramovitch et al., 2025]
Doped $CaCuO_2$, larger U	Static value may appear moderate	Strong ω dependence on phonon scale	Low coherence scale / dynamic local vertex	Counterexample to using only a static correction [Abramovitch et al., 2025]
CoO, polar coupling	Semilocal DFPT reference is qualitatively deficient	DFPT+U changes electronic structure and Fröhlich coupling	Correlated occupation response plus polar screening	Established standard-DFPT limitation [Zhou et al., 2021]
$Ba_{1-x}K_xBiO_3$	DFT linear response	GW perturbation theory enhances coupling	Nonlocal and energy-dependent quasiparticle response	Established beyond-DFT derivative route [Li et al., 2019]

The matrix is deliberately organized by perturbation type rather than by a single material-wide correlation label. Its central unresolved test is whether the operator decomposition predicts the observed contrast without fitting each mode independently.

C Two-day literature and hypothesis-freeze protocol

The initial exploration is limited to two days. Its purpose is not to survey the entire electron–phonon literature, but to decide whether the channel-selective hypothesis is physically defensible and to freeze the first decisive comparison.

Day 1: establish constraints and collect contrasts

1. **Morning: foundations.** Verify the DFPT response convention, physical quasiparticle vertex, proper versus screened vertex, Ward identity, and static/dynamic limit ordering from primary sources [Baroni et al., 2001, Giustino, 2017, Hedin, 1965, Nambu, 1960].
2. **Midday: UEG.** Reproduce the algebra of equations (25) and (26); verify that the complete $K_{\mathrm{total}} \simeq 1$ rather than applying the proper-vertex ratio twice; extract the reported q, r_s, temperature, and error ranges from Cai et al. [2025]; list which statements are numerical and which are exact.
3. **Afternoon: correlated contrasts.** Extract mode definitions, eigenvectors, parameters, normalization, static vertices, and frequency dependence for $SrVO_3$ and $CaCuO_2$.

   The primary source is Abramovitch et al. [2025]. Add CoO and one GWPT case to identify failure mechanisms not represented by local DMFT.
4. **Evening gate.** Produce a one-page evidence table with at least one supporting and one challenging case for each proposed channel.

Day 2: turn the idea into a falsifiable first experiment

1. **Morning: define observables.** Freeze the localized subspace, operator normalization, Fermi-window weights, target g convention, and basis robustness checks.
2. **Midday: compare alternatives.** Test at least three hypotheses on the evidence table: correlation strength alone, quasiparticle residue alone, and channel weight times channel susceptibility. Record which observations each cannot explain.
3. **Afternoon: design the first calculation.** Select a $SrVO_3$ Jahn–Teller/breathing contrast and a UEG density control. Represent the finite-R mode in the phase-correct $2 \times 2 \times 2$ real supercell or equivalent standing-wave basis before assigning channel weights. Specify input data, code path, acceptance thresholds, and the result that would reject the channel hypothesis.
4. **Final gate.** Freeze one paragraph stating the physical hypothesis, one equation for the first correction model, one benchmark table, and one stop/go decision for implementation.

Required outputs after 48 hours

- a source-checked evidence matrix with no mixed normalization conventions;
- a claim ledger labeling exact results, established methods, numerical evidence, and hypotheses;

- the first two material/mode comparisons and their input availability;
- a minimal operator-decomposition prototype specification; and
- an explicit decision: implement, revise the hypothesis, or stop.

The literature sprint is successful only if it reduces uncertainty about the physical hypothesis. The number of papers read is not itself a success metric.

D Claim ledger: what is known and what is proposed

Table 4: Epistemic status of the central statements.

Status	Statement	Main support
ESTABLISHED THEORY	Standard adiabatic DFPT is a self-consistent static Kohn–Sham response and computes the matrix element of $\partial_{\nu\boldsymbol{q}}V_{\mathrm{KS}}$.	Gonze [1995], Baroni et al. [2001], Giustino [2017]
ESTABLISHED	The many-body response closes screening, polarization, self-energy, and a vertex; they are not three arbitrary independent factors.	Hedin [1965]
EXACT LIMIT	A strict unprojected full-space uniform common scalar potential coupled to total charge is an energy-zero shift; the corresponding dynamic Ward limit gives $z\Gamma^0_\omega = 1$.	Nambu [1960] and appendix A
LIMITATION	The static $\boldsymbol{q} \to 0$ density response is not fixed by $z\Gamma = 1$ alone; finite-q density is not a conserved operator.	Ward limit ordering and Fermi-liquid compressibility
EXACT ALGEBRA	In the declared scalar convention, $K_{\mathrm{total}} = z\Gamma_\rho[1-(v+f_{\mathrm{xc}})\chi_s]/[1-vP_{\mathrm{MB}}]$; if both stated matching relations are assumed, substitution gives $K_{\mathrm{total}} = 1$.	equation (25); neither matching relation is an exact constraint and both require numerical evidence
NUMERICAL EVIDENCE	In the tested UEG regime, the complete DFPT-like screened vertex agrees accurately with a high-level vertex over momenta up to $2k_F$.	Cai et al. [2025]; recent preprint
NUMERICAL EVIDENCE	Beyond-DFPT corrections can differ strongly between phonons in the same correlated material and can be frequency dependent.	Abramovitch et al. [2025]
WORKING HYPOTHESIS	The one-body `global_charge`, `site_charge`, `internal`, and `nonlocal` content times channel-resolved many-body amplification is a better predictor of DFPT error than U or z alone; two-body vertices are a separate space.	Proposed here; to be tested on table 2
PROPOSED METHOD	A small static/dynamic operator kernel can correct DFPT with substantially fewer high-level calculations than a full vertex grid.	section 7; not yet validated

Status	Statement	Main support
Open question	The scalar UEG matching condition has a useful, basis-robust matrix generalization in a crystal with local fields and multiple orbitals.	First formal task after the literature sprint

References

Stefano Baroni, Stefano de Gironcoli, Andrea Dal Corso, and Paolo Giannozzi. Phonons and related crystal properties from density-functional perturbation theory. *Reviews of Modern Physics*, 73(2):515–562, 2001. doi: 10.1103/RevModPhys.73.515.

Xavier Gonze. Adiabatic density-functional perturbation theory. *Physical Review A*, 52(2):1096–1114, 1995. doi: 10.1103/PhysRevA.52.1096.

Feliciano Giustino. Electron-phonon interactions from first principles. *Reviews of Modern Physics*, 89(1):015003, 2017. doi: 10.1103/RevModPhys.89.015003.

QuantumBFS. Finding the next-generation microscopic theory and first-principles method for electron-phonon interactions. Quantum Harness Issue 35, 2026. URL `https://github.com/QuantumBFS/quantum.harness/issues/35`. Accessed 30 July 2026.

Andrea Marini, S. Poncé, and X. Gonze. Many-body perturbation theory approach to the electron-phonon interaction with density-functional theory as a starting point. *Physical Review B*, 91 (22):224310, 2015. doi: 10.1103/PhysRevB.91.224310.

Lars Hedin. New method for calculating the one-particle green's function with application to the electron-gas problem. *Physical Review*, 139(3A):A796–A823, 1965. doi: 10.1103/PhysRev.139.A796.

Yoichiro Nambu. Quasi-particles and gauge invariance in the theory of superconductivity. *Physical Review*, 117(3):648–663, 1960. doi: 10.1103/PhysRev.117.648.

Xiansheng Cai, Tao Wang, Shuai Zhang, Tiantian Zhang, Andrew Millis, Boris V. Svistunov, Nikolay V. Prokof'ev, and Kun Chen. Superconductivity in electron liquids: Precision many-body treatment of coulomb interaction. *arXiv preprint arXiv:2512.19382*, 2025. URL `https://arxiv.org/abs/2512.19382`.

David J. Abramovitch, Jennifer Coulter, Sophie Beck, and Andrew Millis. Electron-phonon coupling in correlated metals: A dynamical mean-field theory study. *Physical Review B*, 112(7): 075113, 2025. doi: 10.1103/467t-z5b2.

Jin-Jian Zhou, Jinsoo Park, Iurii Timrov, Andrea Floris, Matteo Cococcioni, Nicola Marzari, and Marco Bernardi. Ab initio electron-phonon interactions in correlated electron systems. *Physical Review Letters*, 127(12):126404, 2021. doi: 10.1103/PhysRevLett.127.126404.

Zhenglu Li, Gabriel Antonius, Meng Wu, Felipe H. da Jornada, and Steven G. Louie. Electron-phonon coupling from ab initio linear-response theory within the GW method: Correlation-enhanced interactions and superconductivity in Ba1-xKxBiO3. *Physical Review Letters*, 122 (18):186402, 2019. doi: 10.1103/PhysRevLett.122.186402.

M. Grilli and C. Castellani. Electron-phonon interactions in the presence of strong correlations. *Physical Review B*, 50(23):16880–16898, 1994. doi: 10.1103/PhysRevB.50.16880.

Yusuke Nomura and Ryotaro Arita. Ab initio downfolding for electron-phonon coupled systems: Constrained density-functional perturbation theory. *Physical Review B*, 92(24):245108, 2015. doi: 10.1103/PhysRevB.92.245108.